

Flashcards

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Q: What are the primary barriers to traditional financial advisory services that FINANCELY seeks to address, and which investor groups are disproportionately affected by these issues?

A: Traditional financial advisory services suffer from accessibility barriers, high costs (e.g., 1-2% advisory fees per year), and fragmented analysis approaches. These issues disproportionately affect retail investors, financially illiterate users, and those from rural or semi-urban regions who often lack formal financial education and are susceptible to misleading promotions.

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Q: Identify and briefly describe the four core analytical modules integrated into FINANCELY, specifying at least one key AI/ML technique or indicator used by each.

A: FINANCELY integrates four core analytical modules: 1) **Sentiment Analysis**, which uses BERT-based models for financial news analysis. 2) **Time Series Forecasting**, employing Facebook Prophet with uncertainty quantification. 3) **Technical Analysis**, utilizing several indicators such as Bollinger Bands, MACD, and RSI. 4) **Fundamental Analysis**, which computes automated Piotroski Scores (ranging from 0-9) based on nine financial criteria.

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Q: Quantify the performance improvements FINANCELY achieved in its experimental results for both price forecasting and sentiment analysis compared to conventional methods.

A: FINANCELY demonstrated a **15% improvement** in price forecasting (Prophet model with 6.18% MAPE) over conventional methods like ARIMA (8.72% MAPE). For sentiment analysis, its BERT-based model achieved **87.3% accuracy**, representing a **14.9% improvement** over lexicon-based methods (72.4% accuracy).

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Q: Explain how the 'Fibonac' AI chatbot enhances FINANCELY's mission of democratizing financial advice, detailing the specific AI technologies it uses for its functionality.

A: The 'Fibonac' AI chatbot enhances FINANCELY's mission by providing individualized financial guidance and educational assistance, simplifying complex signals (like Piotroski Scores or technical indicators) into interpretable explanations for users without financial backgrounds. It leverages **BERT** for query comprehension and intent recognition, and **Google Gemini** for natural language response generation and conversational fluency.

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Q: Beyond providing market insights, what is the broader socio-technical impact FINANCELY aims to achieve, particularly concerning financial literacy and vulnerability?

A: FINANCELY aims to promote financial inclusion and literacy by providing easily accessible, risk-aware advisory support, especially for uninformed, rural, and financially less literate investors. It seeks to lower their vulnerability to deceptive stock promotions, fraud, disinformation, and marketing campaigns by financially troubled companies, thereby democratizing investment intelligence through explainable AI.

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Q: What are two key performance metrics that highlight FINANCELY's efficiency and responsiveness, and what are their values?

A: FINANCELY demonstrates efficiency with an average database query time of 0.08 seconds and a high cache hit rate of 78.5%, indicating effective optimization for repeated analyses. Its average response time is also a quick 1.23 seconds.

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Q: How did FINANCELY's multi-faceted analysis contribute to its successful validation during the NVIDIA (NVDA) case study, and what specific results demonstrated its predictive power?

A: During the NVIDIA (NVDA) case study, FINANCELY successfully integrated multiple analyses: it identified bullish sentiment (+0.72) from AI-related news using a BERT-based module, detected technical buy signals (MACD crossover, RSI support), and predicted price movements within 7% accuracy for a 30-day horizon using Prophet. It also calculated a strong Piotroski score of 8/9, collectively providing comprehensive and interpretable insights.

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Q: Beyond general user satisfaction, what significant societal impact did the user study results suggest for FINANCELY, and what specific data supports this?

A: The user study suggested FINANCELY has significant potential for promoting financial inclusion and literacy in underserved segments. This is supported by the finding that participants from non-finance backgrounds and rural or semi-urban locations reported higher perceived usefulness, alongside an overall 88% of participants reporting improved investment confidence.

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Q: According to Table V, what distinguishes FINANCELY from other major financial advisory systems like Bloomberg Terminal and TradingView in terms of integrated capabilities?

A: Table V shows that FINANCELY uniquely integrates all listed capabilities: Fundamental, Technical, Sentiment, Forecasting, AI Chatbot, Portfolio, and Real-time analysis. In contrast, systems like Bloomberg Terminal lack Sentiment, Forecasting, and AI Chatbot, while TradingView lacks Fundamental, Sentiment, Forecasting, and AI Chatbot, making FINANCELY a more comprehensive solution.

Q: What are two core technological innovations that FINANCELY leverages, and how do they specifically improve upon traditional methods in financial analysis?

A: FINANCELY leverages BERT-based sentiment analysis, achieving 87.3% accuracy and substantially outperforming conventional lexicon-based methods. It also uses Prophet-based forecasting, which provides approximately 15% improvement over traditional ARIMA methods and offers valuable uncertainty intervals for risk-conscious investors, enhancing predictive power and risk assessment.